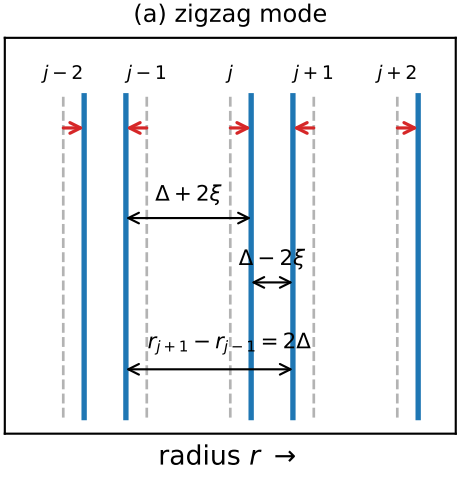
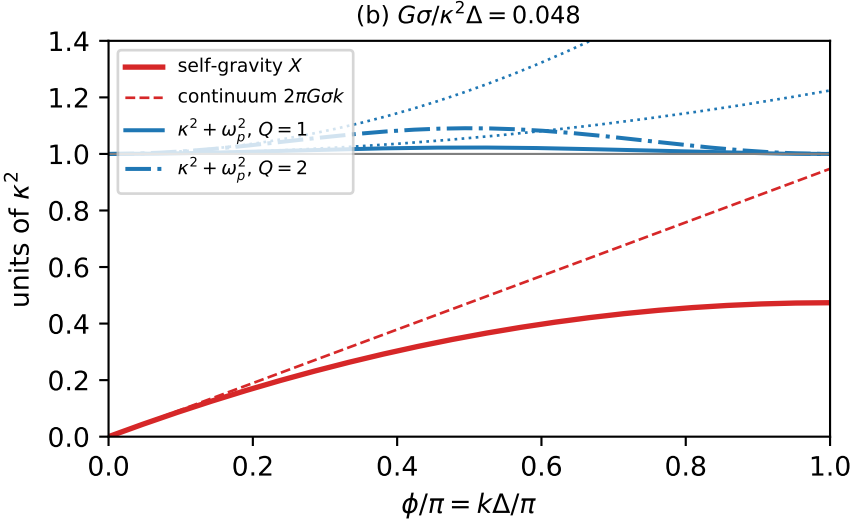

(a) zigzag mode
$j-2$
$j-1$
j
$j+1$
$j+2$
$\Delta + 2\xi$
$\Delta - 2\xi$
$r_{j+1} - r_{j-1} = 2\Delta$
radius r →


(b) $G\sigma/\kappa^2\Delta = 0.048$
self-gravity X
continuum $2\pi G\sigma k$
$\kappa^2 + \omega_\rho^2$, $Q = 1$
$\kappa^2 + \omega_\rho^2$, $Q = 2$
units of κ^2
$\phi/\pi = k\Delta/\pi$
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4